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Photoplethysmography (PPG) Signal Acquisition and Conditioning

Design, Simulation and Analysis of a PPG Circuit

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1 Introduction

1.1 Background

Photoplethysmography (PPG) is a non-invasive optical technique used to detect variations in blood volume within biological tissue. A typical PPG system consists of a light source, usually an LED, and a photodetector such as a photodiode. The light is transmitted into or reflected from the tissue, and the changes in the detected optical intensity are converted into an electrical signal.

The PPG signal contains a small pulsatile component associated with changes in arterial blood volume caused by the cardiac cycle. It is generally superimposed on a much larger slowly varying DC component caused by tissue, venous blood, average blood volume and the optical properties of the measurement site. Consequently, signal conditioning is required before the PPG signal can be reliably observed or processed.

PPG signals are widely used in biomedical instrumentation. Applications include heart-rate monitoring, pulse oximetry, vascular assessment and wearable health-monitoring devices. The simplicity of optical sensing makes PPG particularly suitable for portable and low-cost medical and consumer devices.

1.2 Importance of PPG

The main advantage of PPG is that it allows cardiovascular information to be obtained without an invasive procedure. The periodicity of the pulsatile waveform can be used to estimate heart rate, while the waveform morphology contains information related to blood flow and vascular behaviour.

However, the amplitude of the AC component of a PPG signal is relatively small compared with the DC component. Furthermore, the signal can be affected by ambient light, motion artefacts, power-line interference and electronic noise. Therefore, amplification and filtering are essential parts of a practical PPG instrumentation system.

1.3 Objectives

The main objectives of this practical are:

- To understand the operating principle of a photoplethysmography system.
- To design a circuit capable of detecting the optical variation associated with blood-volume changes.
- To amplify the small electrical signal produced by the photodetector.
- To design suitable high-pass and low-pass filtering stages.
- To reduce unwanted low-frequency drift and high-frequency noise.
- To simulate the circuit using LTspice.
- To observe and analyse the waveform at different stages of the circuit.
- To determine the amplitude and duration of the final PPG waveform.
- To consider practical implementation issues when transferring the circuit to a PCB.

2 Materials and Methods

2.1 Equipment and Components

The following components and equipment were used or considered for the design and simulation of the PPG circuit:

Table 1: Main components used in the PPG signal-conditioning circuit.

Component	Typical value/type	Purpose
LED	Red/IR LED	Optical illumination
Photodiode	Silicon photodiode	Optical detection
Operational amplifier	General-purpose op-amp	Signal amplification
Resistors	Various values	Gain and filter design
Capacitors	Various values	AC coupling and filtering
DC supply	$+V$ / GND	Circuit power supply
LTspice	Simulation software	Circuit simulation

The exact component values should be selected according to the required gain, bandwidth and available supply voltage.

2.2 Reason for Using Each Component

2.2.1 LED

The LED provides optical illumination to the measurement site. When blood volume changes during the cardiac cycle, the amount of light absorbed or reflected by the tissue changes. This produces a corresponding variation in the light received by the photodetector.

2.2.2 Photodiode

The photodiode converts incident optical power into electrical current. The output current contains a large DC component as well as the relatively small pulsatile component associated with the cardiac cycle.

2.2.3 Operational Amplifier

The signal generated by the photodetector is generally too small to be directly used for further processing. An operational amplifier is therefore used to increase the signal amplitude. Additional op-amp stages can be used to implement active filters.

2.2.4 Resistors

Resistors determine amplifier gain, establish bias conditions and form the resistive components of the filter networks.

2.2.5 Capacitors

Capacitors are used for AC coupling and frequency-selective filtering. In particular, a capacitor in combination with a resistor can be used to form a high-pass filter that attenuates the DC component and slow baseline variations.

2.3 Stages of the PPG Circuit

The complete PPG signal-conditioning system can be divided into several functional stages.

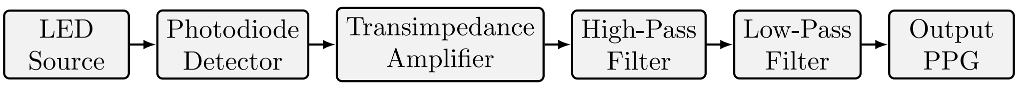

Figure 1: Functional block diagram of the PPG signal-conditioning system.

The main stages are:

1. Optical excitation using an LED.

2. Detection of the reflected or transmitted light using a photodiode.
3. Conversion of the photodiode current into a voltage.
4. Amplification of the small pulsatile signal.
5. High-pass filtering to suppress the DC component and baseline drift.
6. Low-pass filtering to suppress high-frequency noise.
7. Observation of the conditioned PPG waveform.

2.4 Step-by-Step Development

First, the optical sensing arrangement was considered. The LED illuminates the tissue while the photodiode detects the corresponding variation in optical intensity.

Next, the photodiode output was converted into a voltage using an appropriate amplifier configuration. Since the pulsatile signal is small, sufficient gain must be provided while avoiding saturation due to the large DC component.

A high-pass filtering stage was then introduced to reduce the large DC component and slow variations. The signal was subsequently passed through a low-pass filter to attenuate unwanted high-frequency components.

Finally, the complete circuit was simulated in LTspice. The input and output waveforms were observed and the amplitude, period and frequency response of the circuit were analysed.

3 Circuit Diagrams

3.1 Overall Circuit

The complete PPG circuit consists of the optical sensor followed by signal-conditioning stages. A simplified representation is shown below.

$$\boxed{\text{LED}} \rightarrow \boxed{\text{Photodiode}} \rightarrow \boxed{\text{Amplifier}} \rightarrow \boxed{\text{HPF}} \rightarrow \boxed{\text{LPF}} \rightarrow \boxed{\text{Output}}$$

Figure 2: Simplified schematic representation of the PPG system.

3.2 Photodetector and Amplifier Stage

The photodiode generates a current proportional to the incident optical power. The current is converted into a voltage by the amplifier stage.

For a transimpedance amplifier, the approximate output voltage is

$$V_{\text{out}} = -I_{\text{PD}} R_f, \tag{1}$$

where I_{PD} is the photodiode current and R_f is the feedback resistance.

The feedback capacitor, if included, improves stability and limits high-frequency noise.

3.3 High-Pass Filter

The high-pass stage removes the DC component and very slow baseline variations. A first-order RC high-pass filter has a cutoff frequency given by

$$f_L = \frac{1}{2\pi RC}. \tag{2}$$

For example, if

$$R = 100\,\text{k}\Omega$$

and

$$C = 10\,\mu\text{F},$$

then

$$f_L = \frac{1}{2\pi(100 \times 10^3)(10 \times 10^{-6})} \approx 0.159\,\text{Hz}. \tag{3}$$

This allows the normal pulsatile component of a PPG signal to pass while attenuating very slow baseline variations.

3.4 Low-Pass Filter

A low-pass filter is used to attenuate high-frequency noise and unwanted interference. For a first-order RC low-pass filter,

$$f_H = \frac{1}{2\pi RC}. \tag{4}$$

For example, using

$$R = 10\,\text{k}\Omega$$

and

$$C = 1\,\mu\text{F},$$

the cutoff frequency is

$$f_H = \frac{1}{2\pi(10 \times 10^3)(1 \times 10^{-6})} \approx 15.9\,\text{Hz}. \tag{5}$$

The selected upper cutoff should be sufficiently high to preserve the required PPG waveform while reducing high-frequency noise.

3.5 Final Circuit Diagram

The finalized LTspice schematic should be inserted here.

Insert the screenshot of the finalized LTspice circuit here.
Recommended filename: `final_circuit.png`

Figure 3: Finalized PPG circuit implemented in LTspice.

4 Observations

4.1 Input/Photodetector Signal

The signal obtained directly from the photodetector contains both the desired pulsatile component and unwanted components. The DC level is generally much larger than the AC component corresponding to the cardiac pulse.

The simulated photodetector waveform should be inserted below.

Insert LTspice waveform of the photodetector stage here.

Figure 4: Output waveform of the photodetector/amplifier stage.

4.2 High-Pass Filter Output

After high-pass filtering, the large DC component is significantly reduced. The resulting waveform is centred closer to the desired operating level and the pulsatile component becomes easier to observe.

Insert LTspice waveform after the high-pass filter here.

Figure 5: Waveform after high-pass filtering.

4.3 Low-Pass Filter Output

The low-pass stage attenuates high-frequency noise and produces a smoother PPG waveform.

Insert LTspice waveform after the low-pass filter here.

Figure 6: Final filtered PPG waveform.

4.4 Measured Waveform Parameters

The following table should be completed using the actual measurements obtained from LTspice.

Table 2: Measured parameters of the final PPG waveform.

Parameter	Measured value	Unit
Maximum amplitude	**[insert]**	V
Minimum amplitude	**[insert]**	V
Peak-to-peak amplitude	**[insert]**	V
Pulse duration/period	**[insert]**	s
Frequency	**[insert]**	Hz
Estimated heart rate	**[insert]**	BPM

The heart rate can be estimated from the measured pulse period using

$$HR = \frac{60}{T}, \tag{6}$$

where T is the time between two successive PPG peaks in seconds.

5 Analysis

5.1 Amplifier Gain

For a non-inverting operational amplifier, the voltage gain is

$$A_v = 1 + \frac{R_f}{R_g}. \tag{7}$$

For an inverting amplifier, the voltage gain is

$$A_v = -\frac{R_f}{R_{in}}. \tag{8}$$

The appropriate equation should be selected according to the actual amplifier configuration used in the circuit.
For example, if a non-inverting amplifier uses

$$R_f = 100\,\mathrm{k}\Omega$$

and

$$R_g = 10\,\mathrm{k}\Omega,$$

the theoretical voltage gain is

$$A_v = 1 + \frac{100}{10} = 11. \tag{9}$$

The theoretical gain should be compared with the gain obtained from LTspice.

$$A_{v,\text{measured}} = \frac{V_{\text{out}}}{V_{\text{in}}}. \tag{10}$$

5.2 Lower Cutoff Frequency

The lower cutoff frequency is determined mainly by the high-pass filter. For a first-order RC high-pass network,

$$f_L = \frac{1}{2\pi RC}. \tag{11}$$

The cutoff frequency should be selected below the fundamental frequency of the expected PPG signal. This allows the pulsatile component to pass while reducing baseline drift.

For example, for a nominal pulse rate of 60 BPM,

$$f_{\text{pulse}} = \frac{60}{60} = 1\,\text{Hz}. \tag{12}$$

Therefore, a lower cutoff substantially below 1 Hz is desirable for preserving the main pulse waveform.

5.3 Upper Cutoff Frequency

The upper cutoff frequency is determined by the low-pass filter:

$$f_H = \frac{1}{2\pi RC}. \tag{13}$$

The cutoff must be sufficiently high to preserve the important features of the PPG waveform but low enough to attenuate unwanted high-frequency noise.

5.4 Overall Bandwidth

The approximate passband of the complete filtering system is

$$f_L < f < f_H. \tag{14}$$

Thus, the system should be designed such that

$$f_L \ll f_{\text{PPG}} < f_H. \tag{15}$$

For a typical PPG signal, the useful frequency content is concentrated at relatively low frequencies. Therefore, a band-pass response is suitable for removing both slow baseline variations and high-frequency interference.

5.5 Noise Cancellation Using Filtering

The raw PPG signal can contain several unwanted components. These include baseline wander, ambient-light variations, power-line interference and high-frequency electronic noise.

The high-pass filter reduces low-frequency components. In particular, it attenuates the large DC component and slow changes caused by changes in sensor position and tissue properties.

The low-pass filter reduces high-frequency components generated by electronic noise and other interference. The combined response therefore acts approximately as a band-pass filter.

5.6 Bode Plot Analysis

The frequency response of the complete circuit should be obtained using an AC sweep in LTspice. The Bode plot can be used to identify the lower and upper cutoff frequencies.

Insert the LTspice Bode plot of the complete circuit here.

Figure 7: Bode plot of the PPG signal-conditioning circuit.

The magnitude response is expected to show strong attenuation at frequencies below the lower cutoff and above the upper cutoff. The region between these frequencies represents the useful passband of the circuit.

The filtering process can therefore be expressed as

$$H_{\text{total}}(f) = H_{\text{HP}}(f)H_{\text{LP}}(f). \tag{16}$$

For first-order filters,

$$H_{\text{HP}}(f) = \frac{j2\pi fRC}{1 + j2\pi fRC} \tag{17}$$

and

$$H_{\text{LP}}(f) = \frac{1}{1 + j2\pi fRC}. \tag{18}$$

The combined response provides the required band-limiting of the PPG signal.

6 Discussion

6.1 Discussion of Results

The simulated circuit demonstrates the main stages required to obtain a usable PPG signal. The photodetector produces an electrical signal corresponding to variations in detected optical intensity. Since the useful pulsatile component is relatively small, amplification is necessary.

The high-pass filtering stage reduces the large DC component and slow baseline variations. This improves the visibility of the pulsatile signal. The subsequent low-pass filter attenuates high-frequency noise and produces a smoother waveform.

The final waveform should show periodic peaks corresponding to individual cardiac cycles. The interval between consecutive peaks can be used to estimate the pulse rate. The measured values obtained from LTspice should be compared with the theoretical values calculated from the selected circuit parameters.

Any difference between theoretical and simulated results can be attributed to non-ideal operational-amplifier behaviour, component tolerances, loading effects and the characteristics of the simulated photodetector model.

6.2 Practical Implementation Considerations

When implementing the PPG circuit on a PCB, several factors should be considered.

6.2.1 Component Placement

The photodetector section carries a relatively small signal and should be placed close to the first amplification stage. Long PCB tracks can introduce additional noise and unwanted capacitance.

6.2.2 Power Supply Noise

The operational amplifiers should have suitable power-supply decoupling capacitors placed close to their supply pins. Both low-frequency and high-frequency supply noise should be considered.

6.2.3 Grounding

A suitable grounding strategy is important because the detected PPG signal may be much smaller than interference introduced by the surrounding electronics. Ground loops and large current paths through sensitive analogue-ground regions should be avoided.

6.2.4 Ambient Light

The photodetector can respond to unwanted ambient light. The sensor should therefore be physically shielded from external light where possible. Optical shielding and appropriate mechanical positioning can significantly improve the signal-to-noise ratio.

6.2.5 Motion Artefacts

Movement of the sensor relative to the skin can produce changes in the detected optical signal that are unrelated to the cardiac cycle. The sensor should therefore be mounted securely while avoiding excessive pressure on the measurement site.

6.2.6 PCB Layout

Sensitive analogue signal tracks should be kept short. High-speed digital signals, clock lines and switching power-supply tracks should be kept away from the photodetector and amplifier input.

6.2.7 Component Tolerances

The actual cutoff frequencies depend on the tolerances of the resistors and capacitors. Components with appropriate tolerances should therefore be selected when accurate filter characteristics are required.

6.2.8 Operational Amplifier Selection

The operational amplifier should have adequate bandwidth, input noise performance, input bias characteristics and output swing for the intended circuit. The available supply voltage must also be compatible with the selected device.

6.3 Limitations

The simulated circuit does not fully reproduce all the effects present in a real PPG measurement. Practical measurements are affected by skin properties, sensor positioning, ambient light, motion, photodiode characteristics and electrical interference.

Therefore, a successful LTspice simulation does not by itself guarantee an equally clean waveform in hardware. Experimental validation is required after PCB implementation.

7 Conclusion

A photoplethysmography signal-acquisition and conditioning system was studied and simulated. The system consists of an optical source, photodetector, amplification stage, high-pass filter and low-pass filter. Each stage has a specific function in converting the small optical variation associated with the cardiac cycle into a measurable electrical waveform.

The amplification stage increases the magnitude of the detected signal, while the high-pass filter reduces the large DC component and low-frequency baseline variations. The low-pass filter attenuates high-frequency noise. Consequently, the combined filtering stages produce a band-limited signal in which the pulsatile component is easier to observe.

The theoretical gain and cutoff frequencies can be calculated from the selected resistor and capacitor values and compared with the LTspice simulation results. The final PPG waveform can then be analysed to determine its amplitude, period and corresponding heart rate.

For future improvement, a higher-order active band-pass filter could be implemented to provide sharper attenuation outside the desired frequency range. Shielding against ambient light, improved PCB grounding, low-noise amplification and mechanical stabilization of the sensor could further improve the signal-to-noise ratio. Additional improvements could include automatic gain control, digital filtering and motion-artifact rejection.

Overall, the practical demonstrates the importance of appropriate amplification and frequency-selective filtering in biomedical instrumentation and provides a foundation for developing a practical PPG-based heart-rate monitoring system.

References

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